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Green Team Documentation

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The Definition of Terms & Abbreviations

AD	Active Directory
FTP	File Transfer Protocol
VNC	Virtual Network Computing
IP	Internet Protocol
WAN	Wide Area Network
LAN	Local Area Network
NAT	Network Address Translation
RDP	Remote Desktop Protocol
DNS	Domain Name System
IDS	Intrusion Detection System
IPS	Intrusion Prevention System
NSM	Network Security Monitoring
IPv4	Internet Protocol version 4
SSH	Secure Shell
HTTP	Hypertext Transfer Protocol
IT	Information Technology
OU	Organizational Unit
SSSD	System Security Services Daemon
GPO	Group Policy Object

Introduction

This document provides a comprehensive, step-by-step guide outlining how users can connect to the network and utilize its available services. It also offers an overview of how the underlying systems and servers function together, highlighting the internal structure and communication within the network environment. By following the instructions detailed in this guide, users will be able to successfully authenticate and access organizational resources in accordance with their assigned roles and permissions.

1.Public Services

The public services that are accessible for the users and their domain and server names are mentioned on the table below:

Domain Name	Server Name
ad.team4.isucdc.com	Active Directory
admin.team4.isucdc.com	Admin User Interface
tickets.team4.isucdc.com	Ticketing Machine
www.team4.isucdc.com	Web Application

2. Servers

This section delivers a detailed explanation of the purpose of each server and outlines how permissions and access levels are configured according to individual user roles.

2.1 Active Directory (AD)

Active Directory is a fundamental component of our server environment, as it consolidates the management of user accounts, security configurations, and access rights throughout the network. It serves as the core system for authentication and authorization, enabling users to securely connect to the resources and services required for their work.

Through centralized control and structured organization, Active Directory enhances administrative efficiency, improves access management, and reinforces overall network protection. This structured approach ensures consistent security policies, simplifies account administration, and supports a stable and well-managed IT infrastructure.

2.1.1 Remote Desktop Connection

This service is intended for authorized IT Administrators responsible for maintaining and securing the organization's infrastructure.

Remote Desktop Protocol (RDP) enables secure remote management of the Active Directory server, allowing administrators to perform system maintenance, configuration updates, and security monitoring as required by operational needs.

1. Open Remote Desktop Connection service, and fill in the computer and username fields. Click connect once all fields are filled properly.
2. Afterwards, give your credentials. Once you fill out the field, click OK.
3. In case a warning pop-up window appears, click Yes to continue.
4. After all steps are completed correctly, you will be connected to AD.

2.1.2 Managing Users in Active Directory

1. Open the Server Manager as shown below:

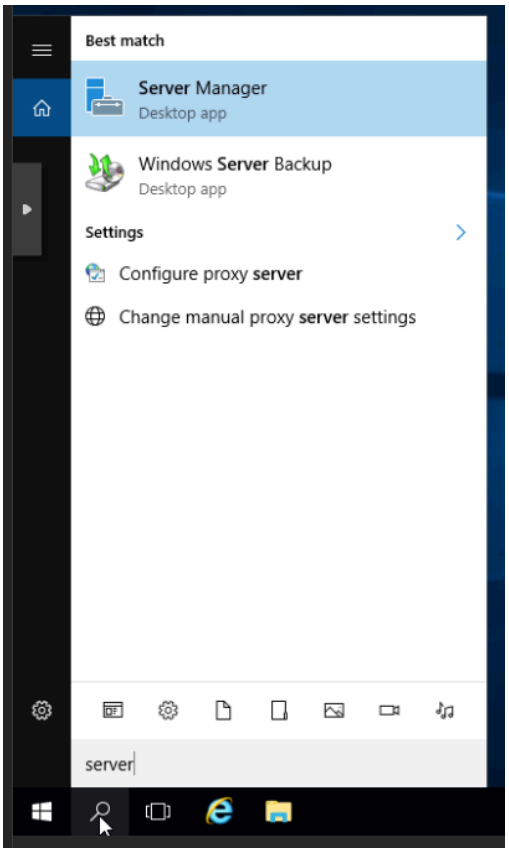

Figure 1. Accessing the Server Manager Dashboard

2. Once open, Server Manager allows you to add, remove, and manage all user accounts as shown below:

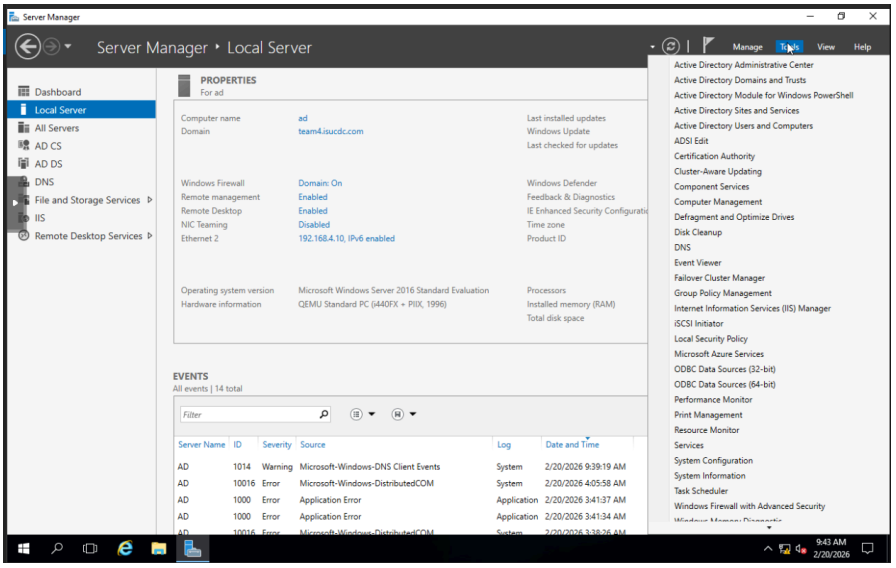

Figure 2. Server Manager Local Server Interface

3. In order for you to add Domain users, navigate to Tools tab, and choose Active Directory Users and Computers (as shown in the prior image). After that, it will open the tab that allows you to view all users. In our case, if we wish to add more Domain users, clicking on the user profile icon will allow us to do so, as shown in the following image:

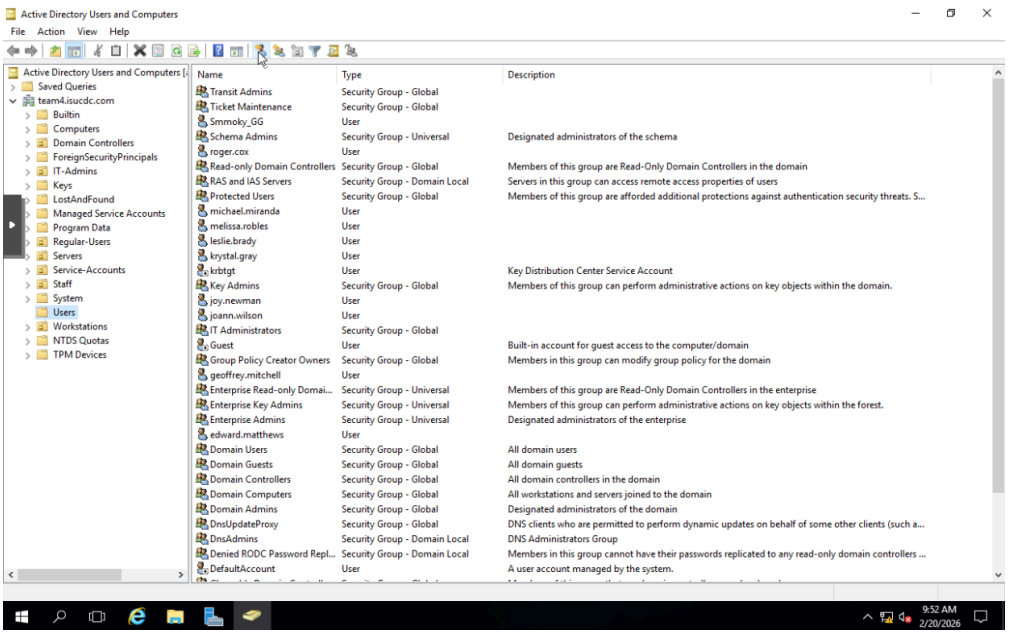

Figure 3. Viewing Domain Users and Groups

4. Afterwards, fill in all the required information for the user such as First name, Last name, and Username, in the dialogue box (shown in the image below):

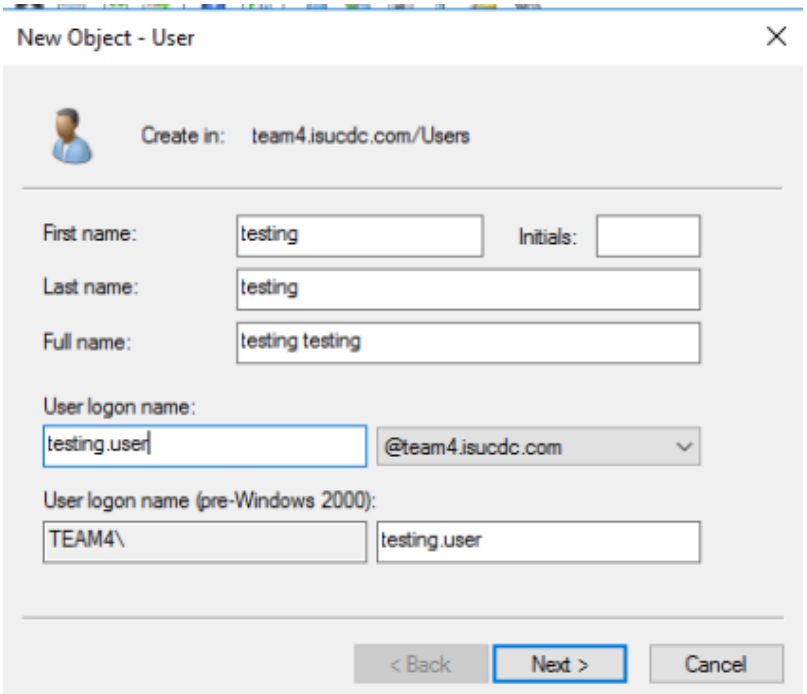

Figure 4. Entering User Information and Logon Name

5. Once all required fields are filled, click on the Next button, which leads you to the password window (with the 'User must change password at next logon' option already enabled by default):

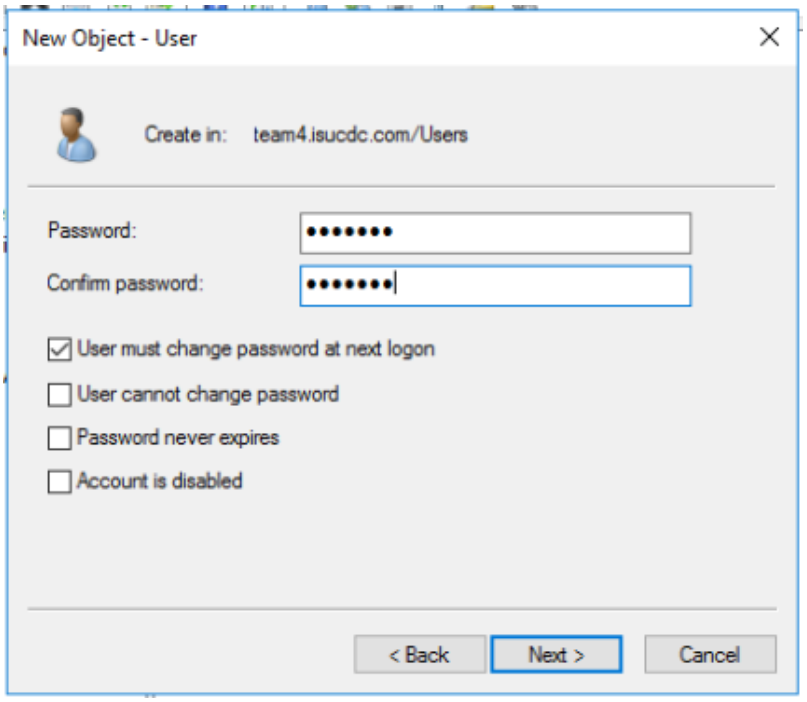

Figure 5. Setting User Password and Security Options

6. Click on the Next button again, and you will be presented with the information about the user you are creating. By clicking on the Finish button, it will create the new user:

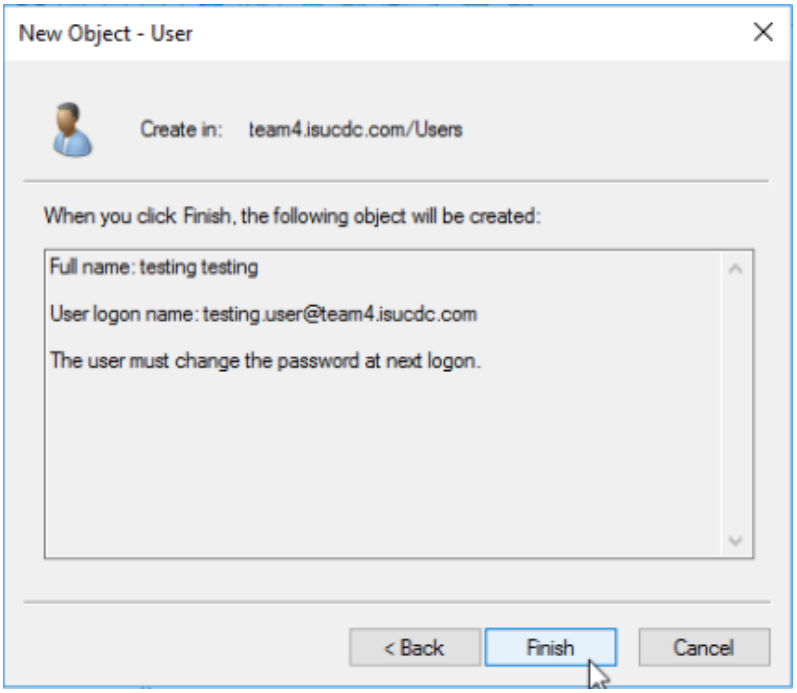

Figure 6. Completing the User Creation Process

2.2 Admin User Interface

This system is intended for authorized Transit Administrators only. It is not intended for public users.

2.2.1 Purpose of the System

The Admin User Interface is the city's administrative portal used to monitor and manage the transportation system. It allows authorized Transit Administrators to view vehicle status, manage fleet operations, and oversee system activity.

2.2.2 How to Access the Admin User Interface

1. Open a web browser (Google Chrome, Microsoft Edge, etc.).
2. In the address bar, enter: http://admin.team4.isucdc.com.
3. After pressing Enter, the Login Page should appear:

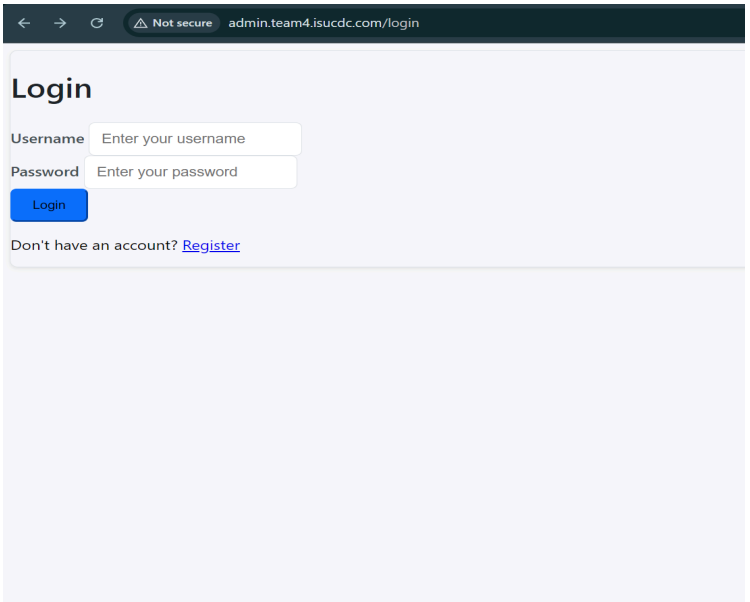

Figure 7. Verifying Domain Account Access Completing the User Creation Process

2.3 Ticketing Machine

2.3.1 General System Overview

The Ticketing Machine is a cornerstone of the Central Downtown City (CDC) public transit infrastructure. Its primary functions and roles are outlined below:

- **Core Purpose**: This system is responsible for the digital lifecycle of transit passes, specifically the issuing and validation of tickets for all city bus and train services.
- **System Representation**: In the broader network, this machine serves as the digital backend for the physical ticket readers located at transit stations and onboard vehicles.
- **Operational Impact**: Because thousands of residents depend on these systems to commute to work and school, the availability of this machine is essential for maintaining public order and safety.
- **Financial Integrity**: The system ensures the financial health of the CDC by preventing "free rides" and ensuring that every passenger has a valid, paid-for ticket before boarding.
- **Service Integration**: It works in tandem with the Active Directory for authentication and the Management Backend to ensure that ticket payments and fleet data are synchronized.

2.3.2 User Access & Authentication

Access to the Ticketing Machine is strictly managed to ensure the reliability of the city's transit services. To perform routine maintenance or system updates, authorized users such as IT Administrators and Ticket Maintenance staff must connect to the server using

Remote Desktop Protocol (RDP). This connection is established through port 3389 and should be directed to the system's dedicated hostname: tickets.team4.isucdc.com.

When logging in, staff should utilize the standard administrative credentials: the default username is Administrator and the default password is cdc. It is critical that all maintenance personnel are assigned Administrator-level account privileges; without these permissions, staff will be unable to manage the ticketing application or execute necessary system tasks.

For security and operational integrity, RDP access is only available from within the competition network. Additionally, because the Ticketing Machine is integrated with the Central Downtown City Active Directory (ad.team4.isucdc.com), user authentication is synchronized across the entire transit network to provide a seamless login experience for our maintenance teams.

2.4 Web Application

The **CDC Public Website** is the primary digital gateway for our passengers. It is designed to be mobile-friendly and accessible to all residents, providing the tools needed to navigate the city's transit network effectively.

2.4.1 How to Access the Web Application

1. Open your web browser (Google Chrome, Microsoft Edge, etc.).
2. In the address bar, enter: http://www.team4.isucdc.com.
3. After pressing Enter, the following Home Page should appear:

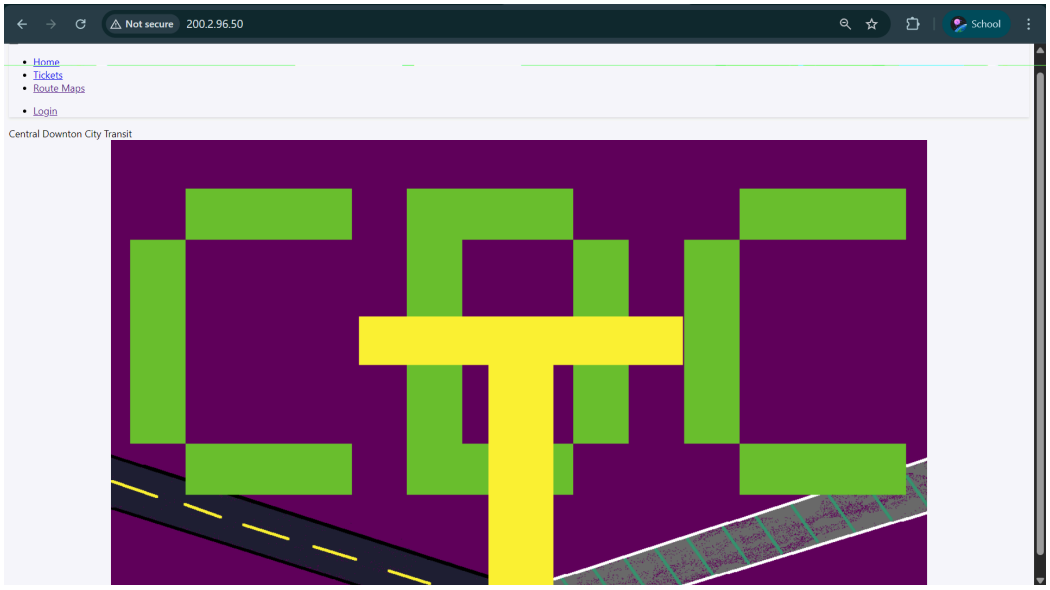

Figure 8. Central Downtown City Transit Home Screen

4. From here, you can access the Tickets Tab, Route Maps Tab, and the Login Tab, all of which are shown below:

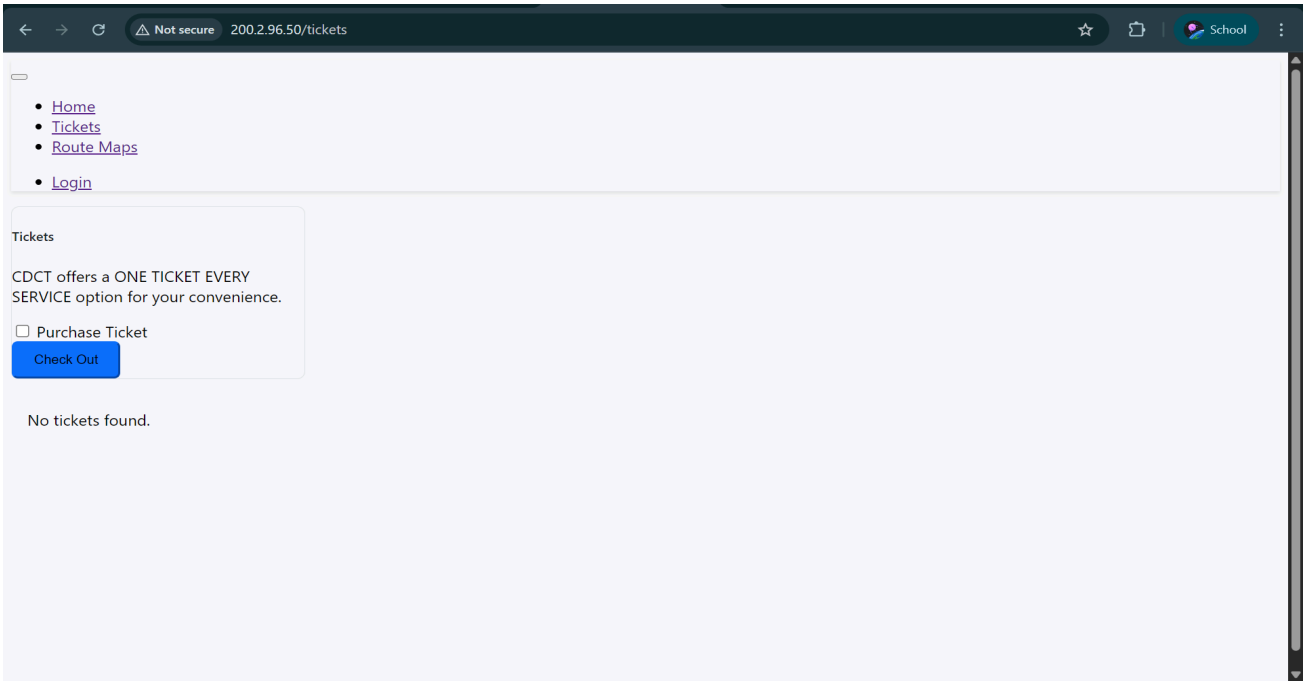

Figure 9. Central Downtown City Transit Home Screen

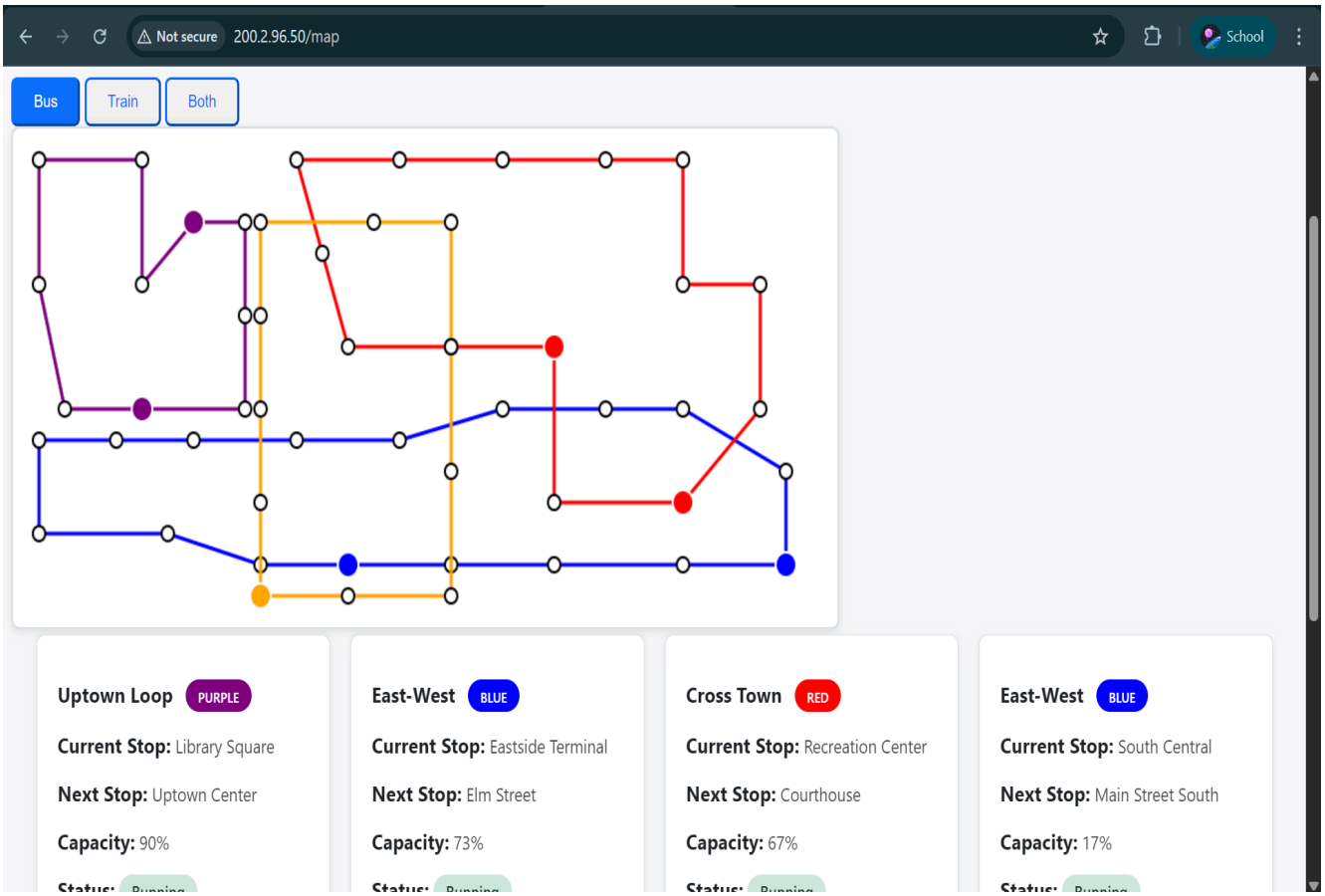

Figure 10. Route Maps Visualization and Service Status

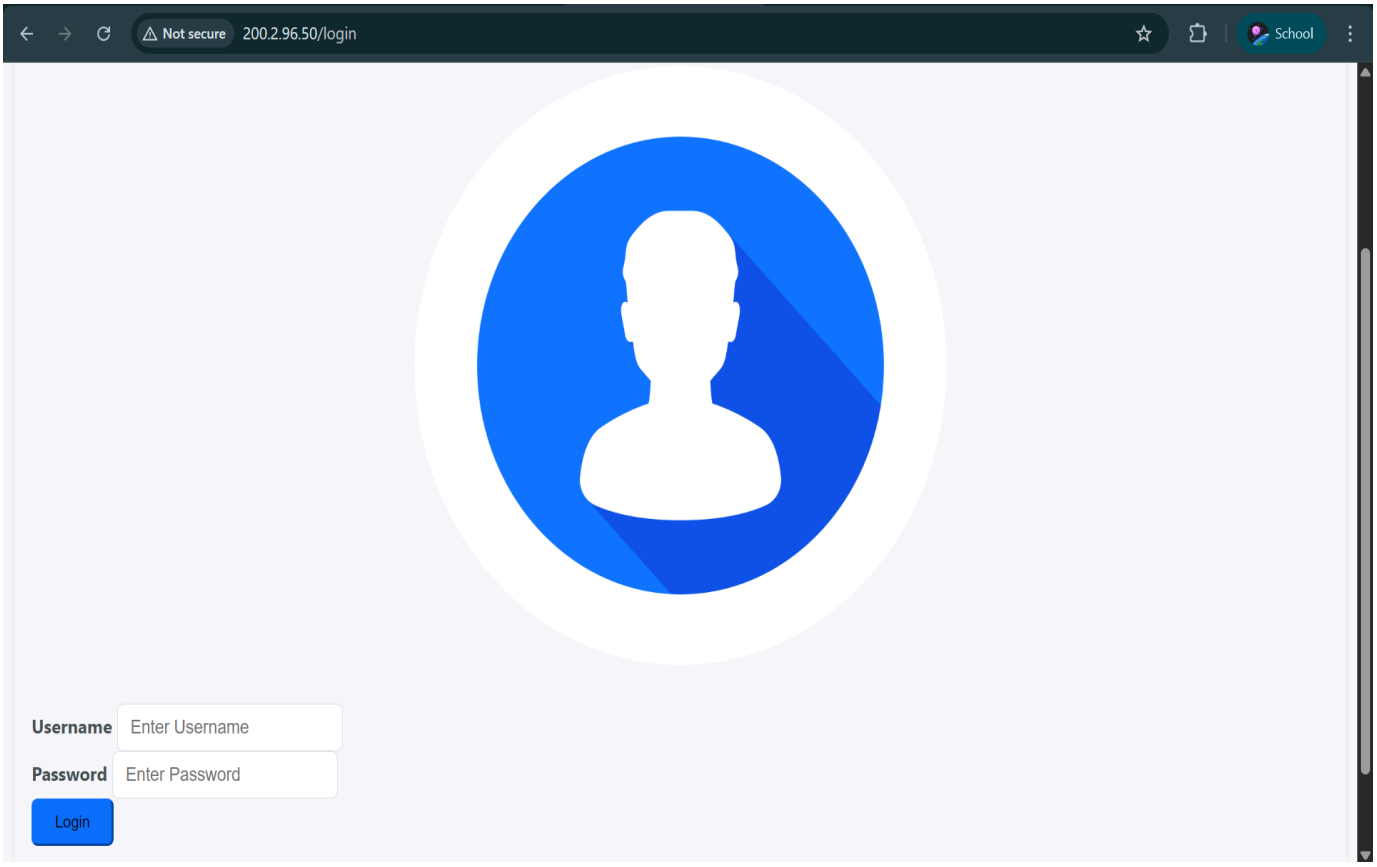

Figure 11. Standard User Authentication Screen